LAB 3 HOMEWORK REPORT

Course: Database and Web Application Development
Lecturers: Nguyen Thanh Binh, Nguyen Phan Thao Nguyen

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Github: https://github.com/onecot/csbu109-labs/tree/main/lab3/homework

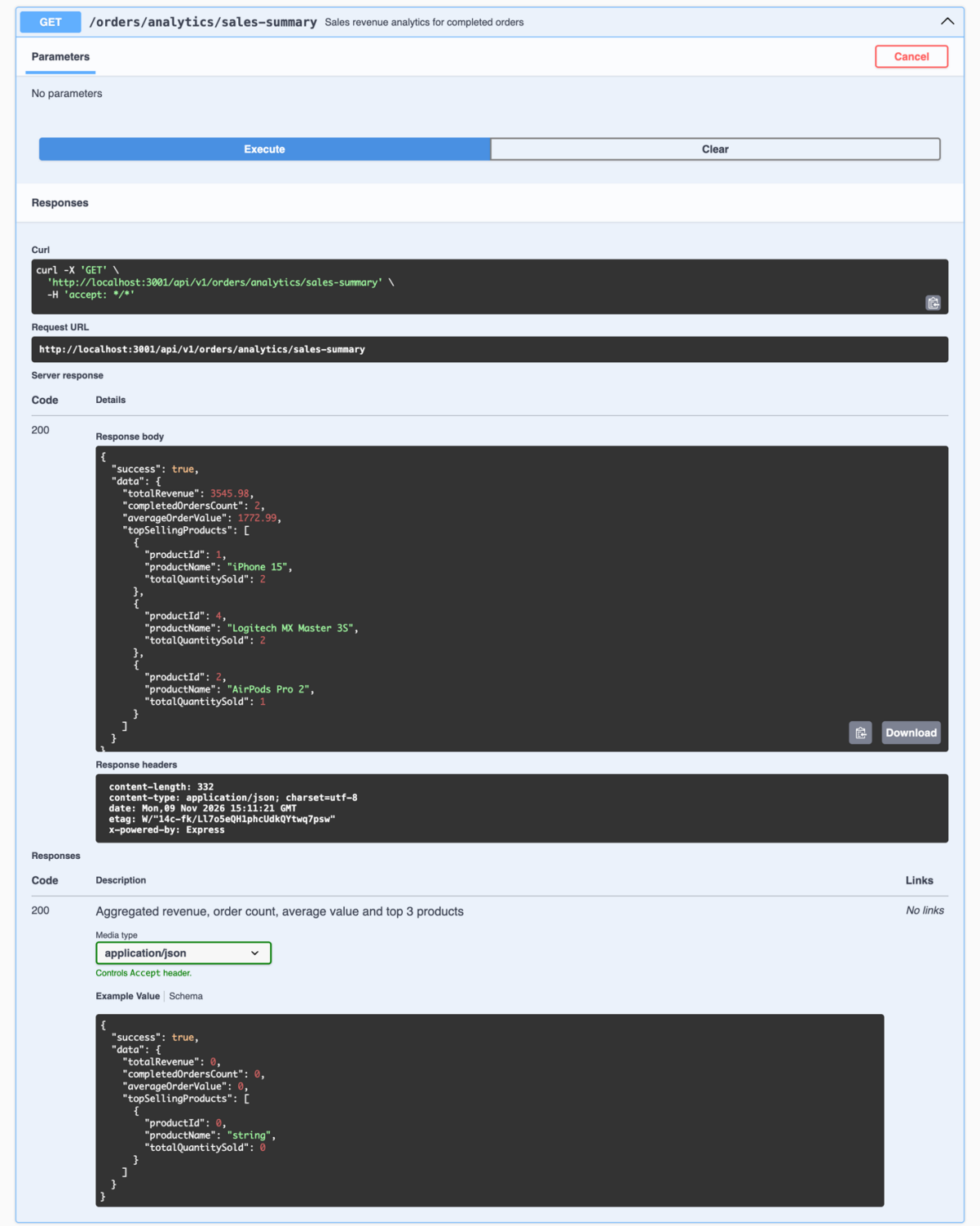
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Homework: Advanced Orders & Inventory Management API with MySQL

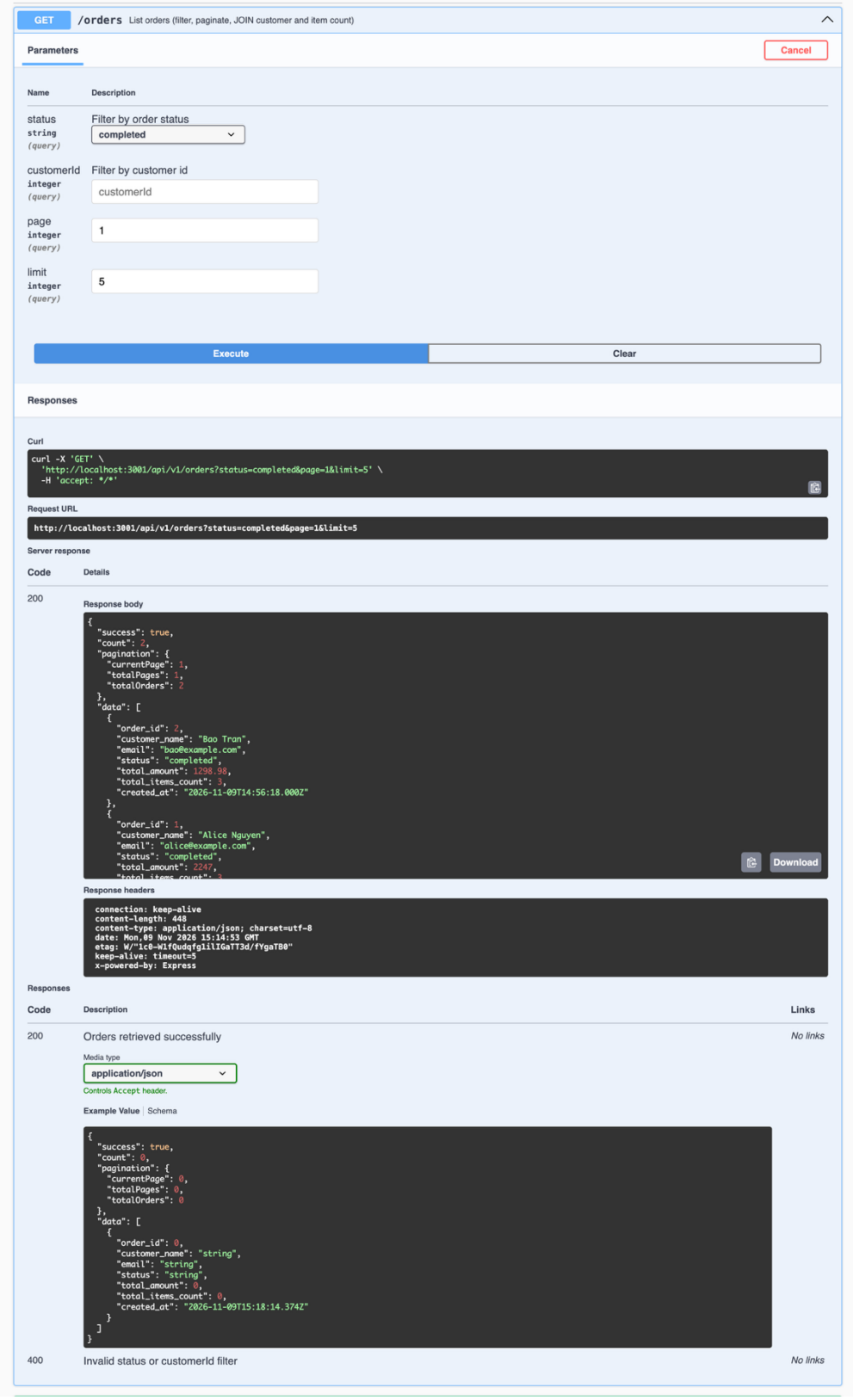
1. Sales summary

GET /orders/analytics/sales-summary



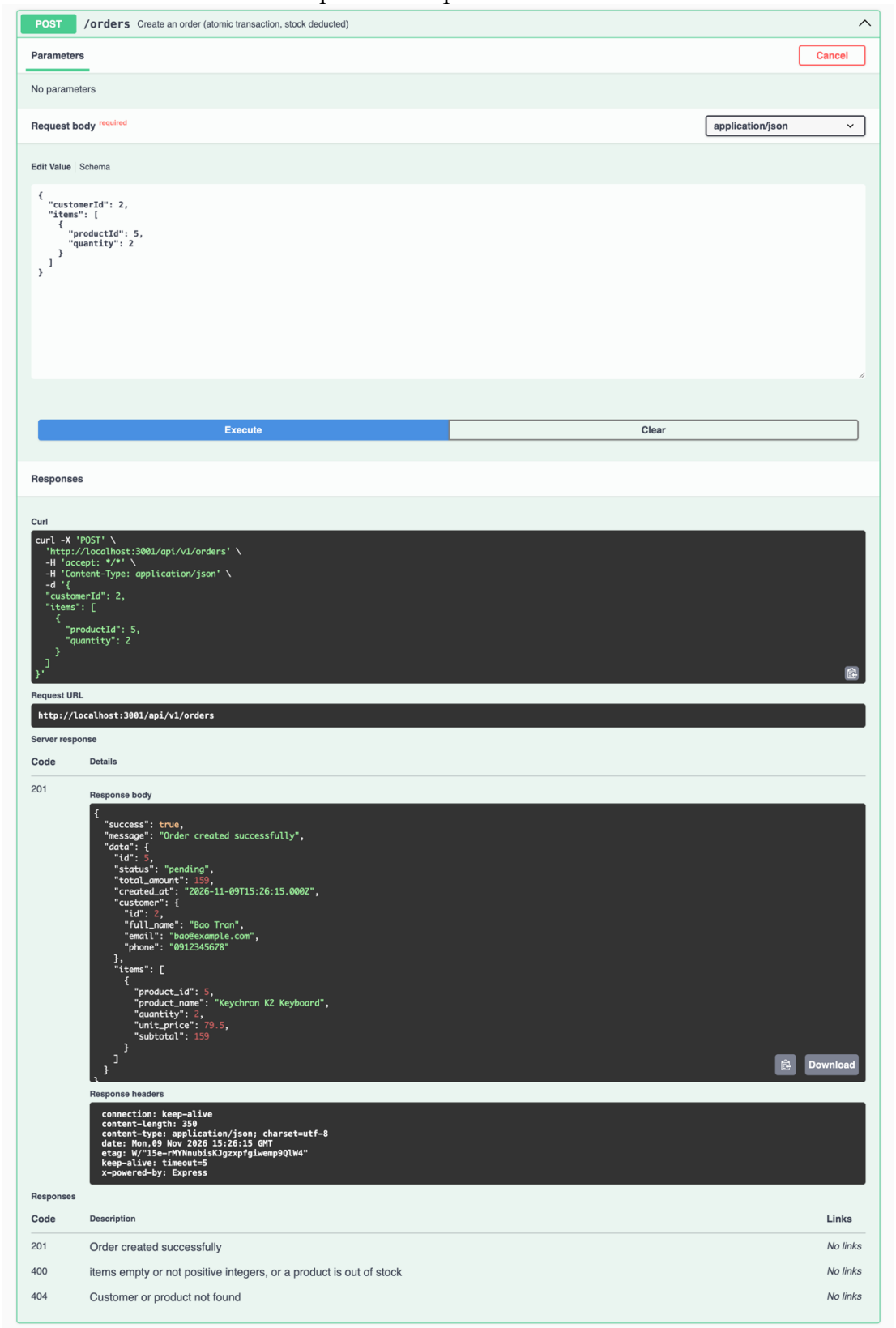
2. Filtered order list

GET /orders?status=completed&page=1&limit=5



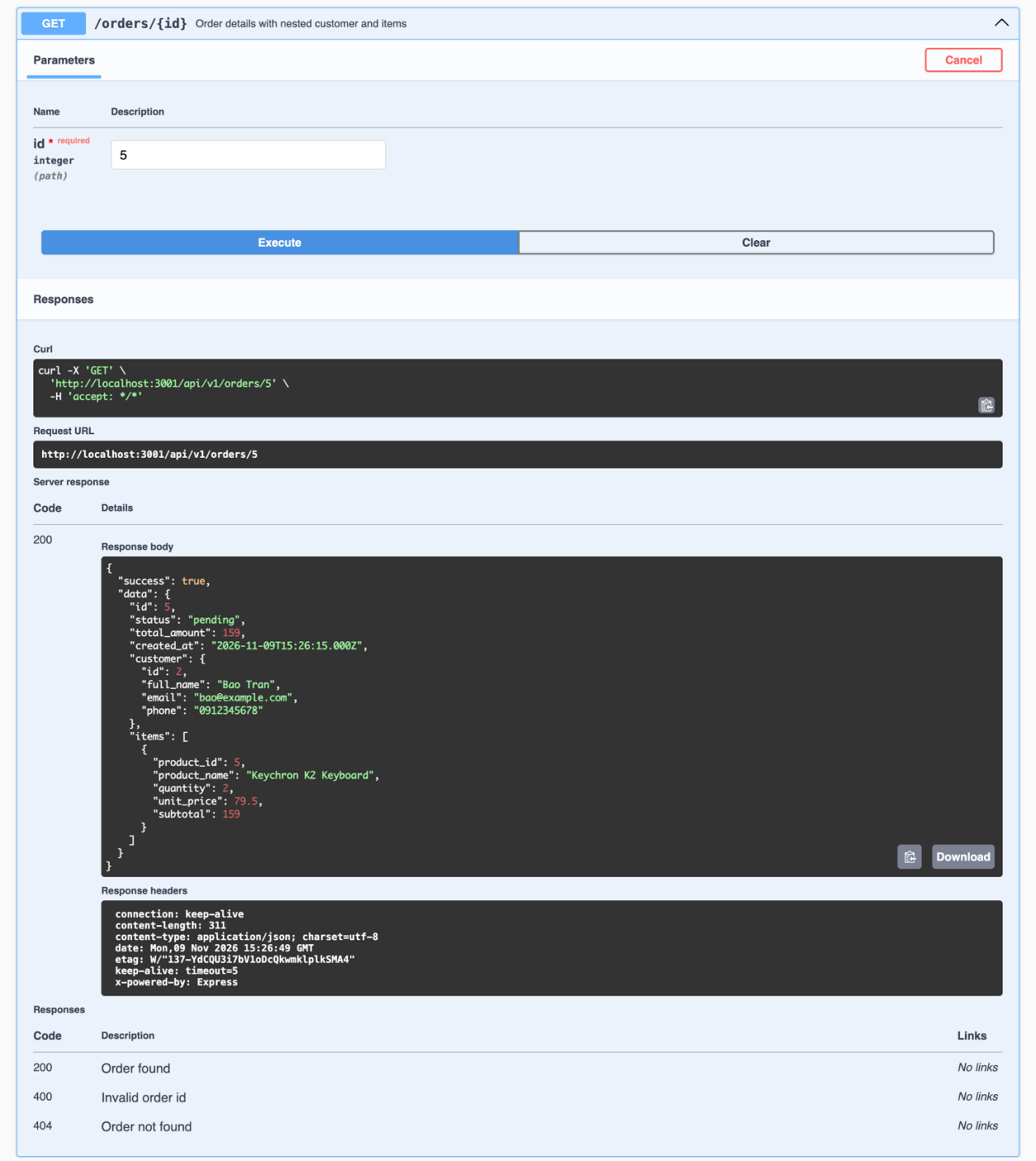
3. Order creation

POST /orders to create order with 2 products of productId=5



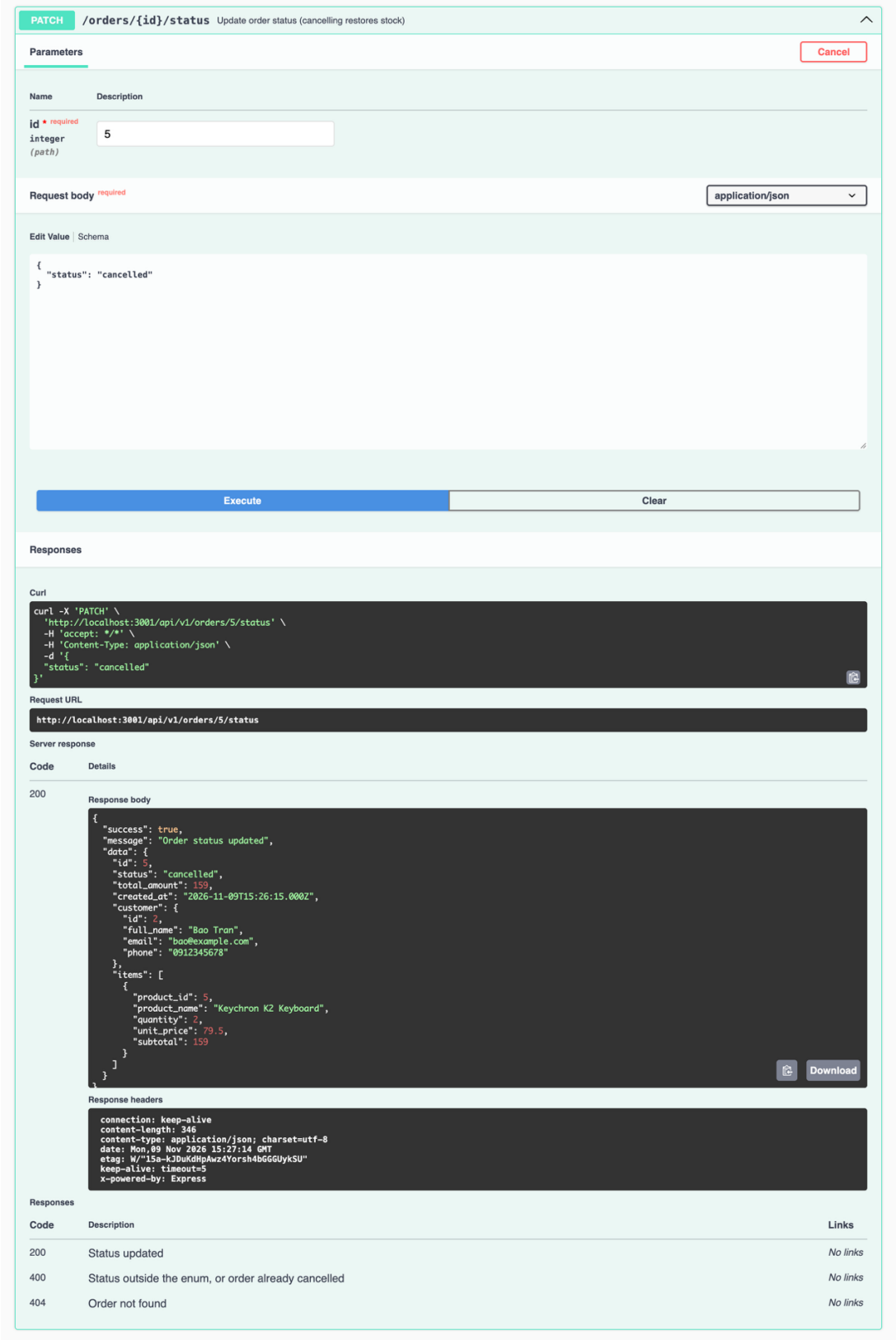
4. Order detail

GET /orders/5



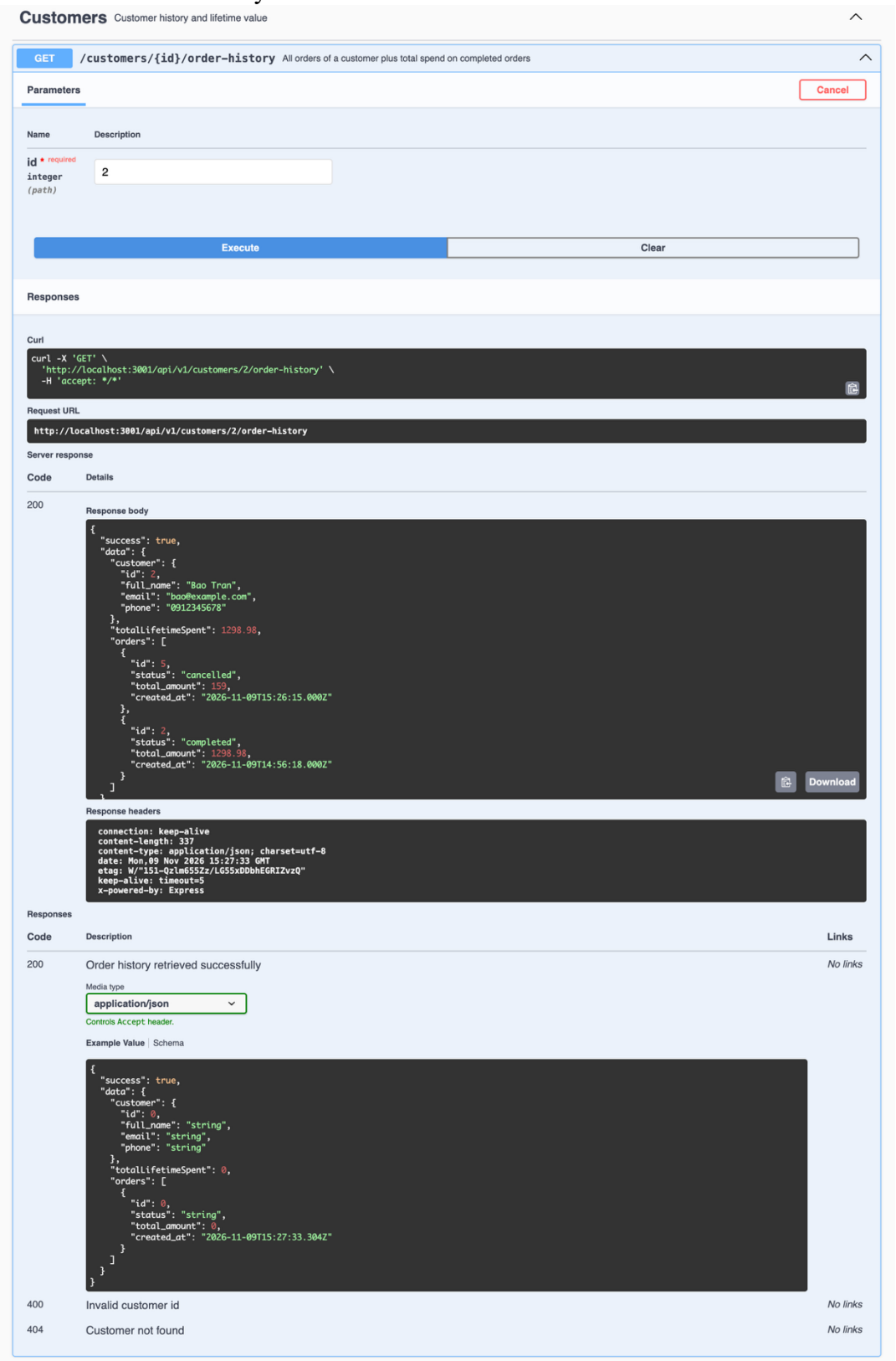
5. Status change

PATCH /orders/5/status



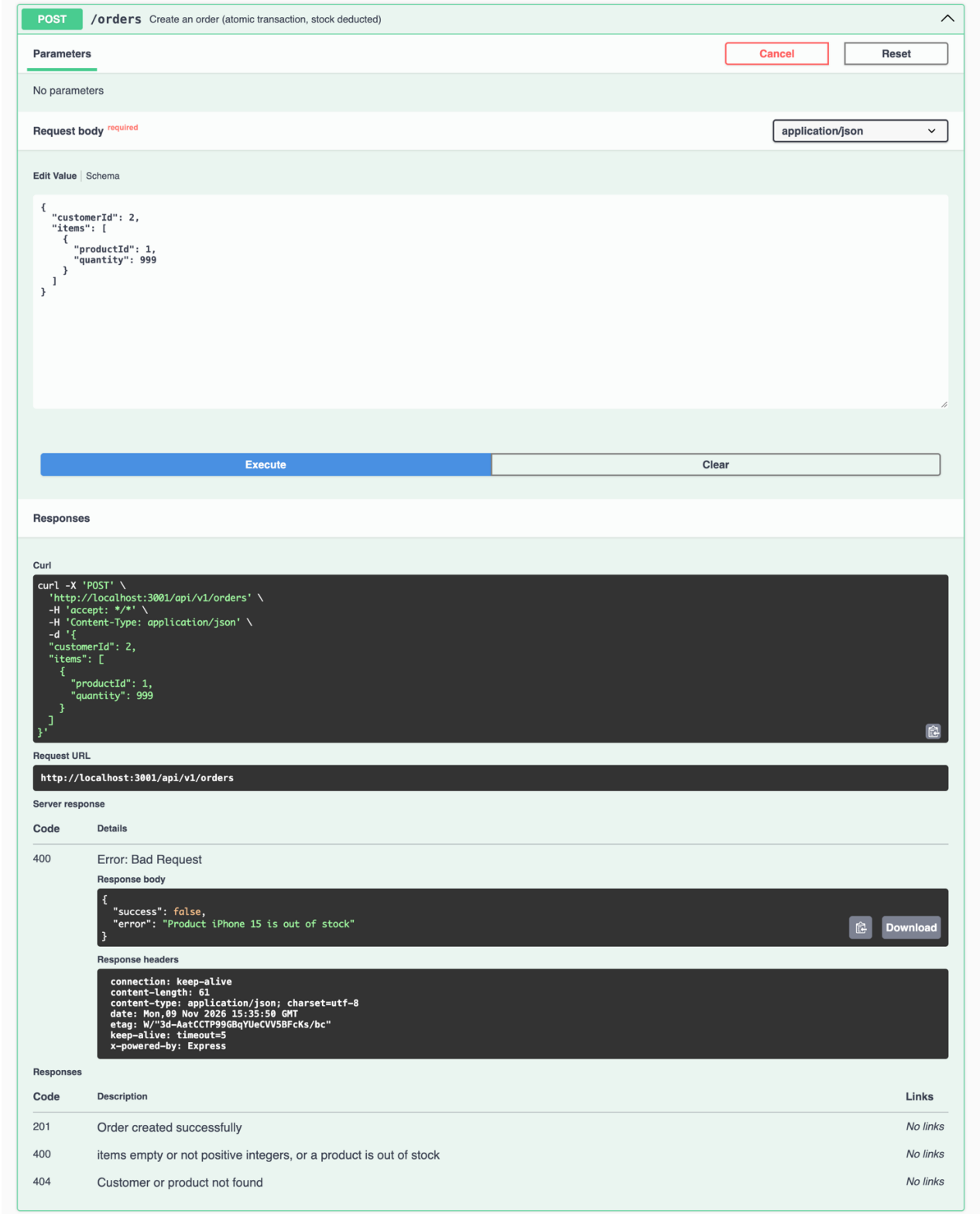
6. Customer history

GET /customers/2/order-history



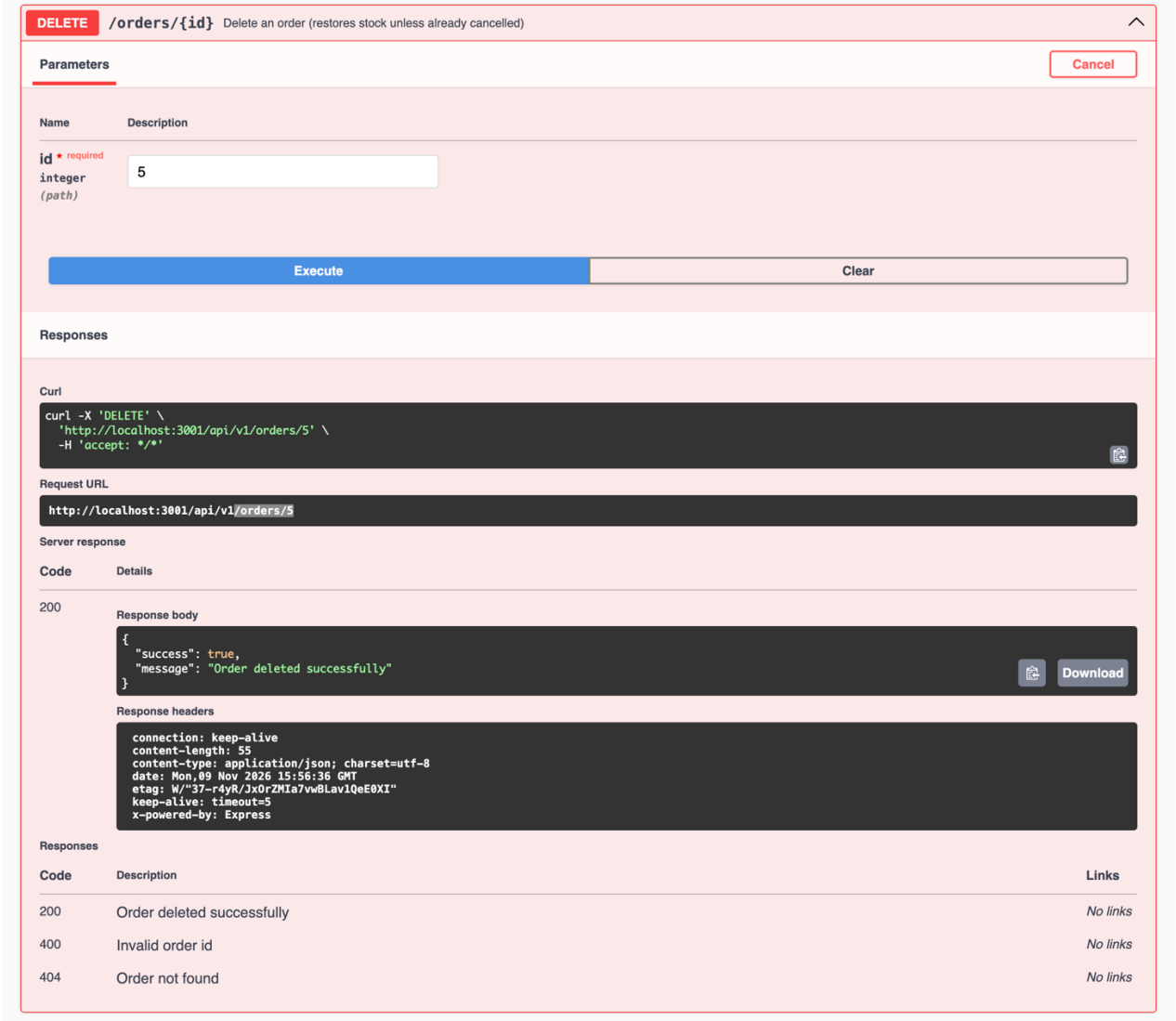
7. Insufficient stock

POST /orders to create order with 999 products of productId=1

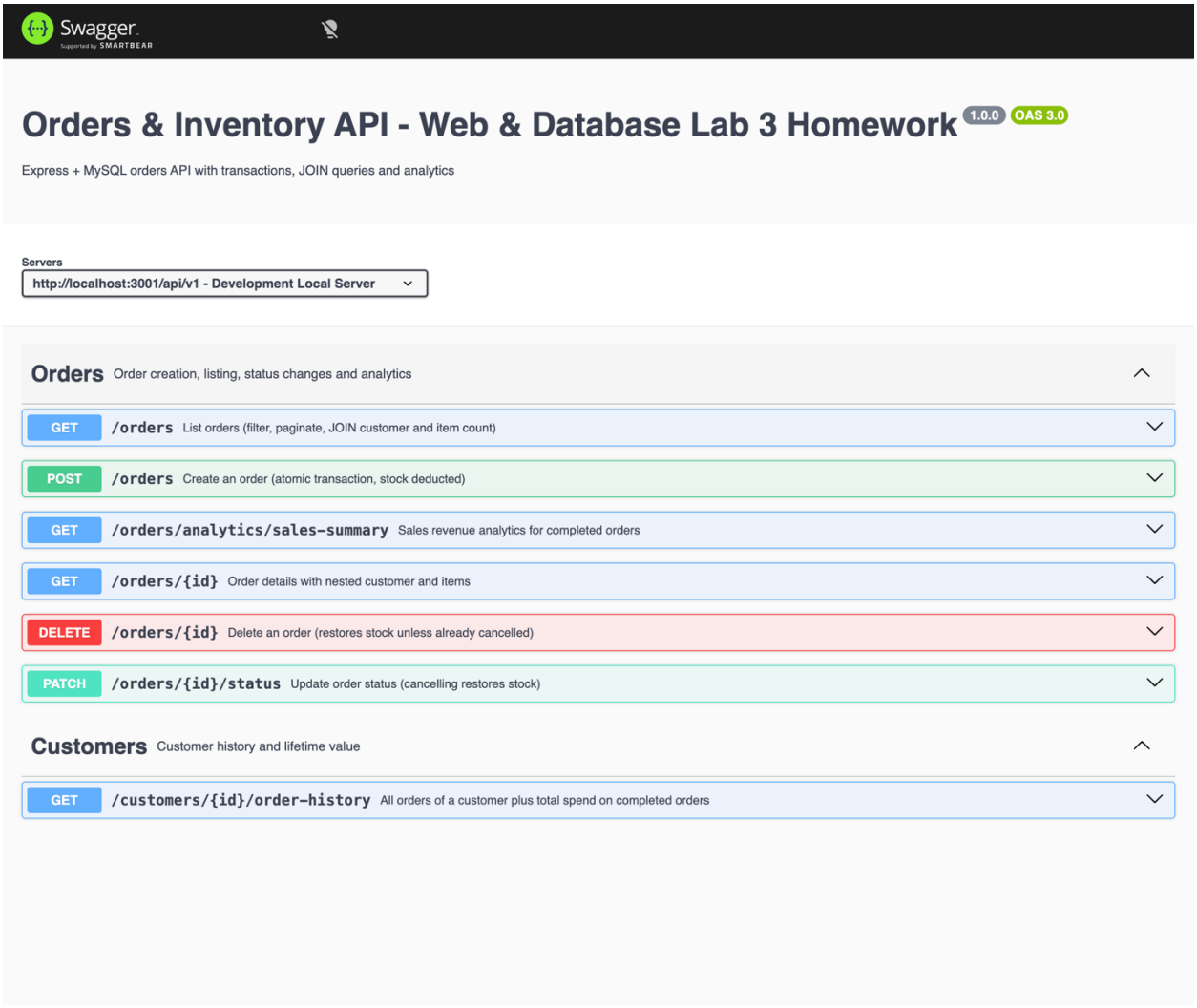


8. Order deletion

DELETE /orders/5



9. Swagger UI


Swagger
Supported by SMARTBEAR
Orders & Inventory API - Web & Database Lab 3 Homework
1.0.0
OAS 3.0
Express + MySQL orders API with transactions, JOIN queries and analytics
Servers
http://localhost:3001/api/v1 - Development Local Server
Orders
Order creation, listing, status changes and analytics
GET /orders List orders (filter, paginate, JOIN customer and item count)
POST /orders Create an order (atomic transaction, stock deducted)
GET /orders/analytics/sales-summary Sales revenue analytics for completed orders
GET /orders/{id} Order details with nested customer and items
DELETE /orders/{id} Delete an order (restores stock unless already cancelled)
PATCH /orders/{id}/status Update order status (cancelling restores stock)
Customers
Customer history and lifetime value
GET /customers/{id}/order-history All orders of a customer plus total spend on completed orders